

# Reference-Conditioned Structural Assessment of Low-Resource Chinese Calligraphy with Overlapping Stroke Parsing and Human-Audited Spatial Scoring

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## Abstract

**Purpose:** Two handwritten instances of a Chinese character can be equally legible yet differ in local stroke structure. We formulate reference-conditioned structural assessment as an auditable document-analysis task under limited labelled data.

**Methods:** A six-channel parser predicts five overlapping direction masks and an endpoint mask. U-Net, DeepLabV3+, and SegFormer-B2 are compared over three seeds on standard and character-disjoint splits of 769 quality-controlled samples. We evaluate constrained similarity registration and diagnostic evidence on cached masks from 200 references under controlled perturbations. Three structural scores are assessed on 150 natural pairs rated blindly by three domain-qualified raters, two of them authors.

**Results:** DeepLabV3+ leads standard-split direction Macro Dice ( $0.9630 \pm 0.0006$ ); SegFormer-B2 leads character-disjoint direction parsing ( $0.8866 \pm 0.0008$ ); U-Net leads strict endpoint F1. Registration reduces nuisance penalties by 30.118 points relative to no alignment on clipping-safe cases while increasing structural penalties by 11.189 points. The frozen production score, a coverage-aware score, and a retrospectively developed aligned spatial-distribution similarity (ASDS) reach Spearman correlations of 0.297, 0.429, and 0.556, respectively. Character-grouped out-of-fold ASDS retains 0.540; direct-ink ASDS shows nearly identical association ( $\Delta\rho = 0.0004$ ). Localized evidence reaches Recall@3 of 0.701 but exact-cell localization of 0.109.

**Conclusion:** The framework separates scalar structural agreement from direction-specific diagnostic evidence, with explicit limits on generalization, human validity, and localization.

**Keywords:** Chinese calligraphy, document analysis, multi-label segmentation, reference-based assessment, human validation, explainable feedback

## 1 Introduction

Calligraphy—the writing of Chinese characters with a brush—is a foundational form of traditional Chinese culture. Its principal script styles, seal (*zhuan*), clerical (*li*), regular (*kai*), running (*xing*),

and cursive (*cao*), employ distinct stroke techniques and convey a wide range of qualities, from dignified and elegant to vigorous, unrestrained, archaic, and lively. Two handwritten instances of the same Chinese character can be equally legible

yet differ in the local structure relevant to calligraphy (*shufa*) instruction. A teacher may identify a shortened horizontal stroke, a displaced hook, a low component, or an endpoint that fails to meet its neighbour. Figure 1(a) illustrates structural differences using a regular-script reference and a running-script candidate of the same character. Both are recognizable, but their stroke proportions, junctions, and component spacing differ. These cross-style differences illustrate the comparison task; they do not establish that either instance is incorrect. Useful feedback must identify the relevant discrepancies and their locations relative to the selected reference.

Global placement can vary without changing the local structure being assessed. A character may be slightly larger, shifted on the canvas, or tilted by a degree or two. Missing terminal fragments, over-wide strokes, and misproportioned components should remain visible after such differences are compensated. Reference-conditioned assessment must therefore separate global placement from local discrepancy and return inspectable evidence alongside a scalar score.

Recognition, style classification, and generation address character identity, style category, and glyph synthesis, respectively [14, 25, 26]. These objectives do not directly evaluate spatially resolved, direction-specific discrepancies between two instances. Formulating same-character structural comparison as a document-analysis task exposes four computer-vision challenges.

- C1 *Overlapping labels under scarce annotation.* At crossings and adhesions, a pixel can carry multiple direction labels, which a mutually exclusive partition cannot represent. Endpoint evidence is sparse, and dense annotation is limited to 769 quality-controlled observations of 40 character identities, without reliable writer identities.
- C2 *Global nuisance versus local structure.* Direct overlap confounds structure with translation, scale, and rotation. Flexible registration can erase proportion or component-placement errors, while an overlap score can reward direction channels that are empty in both images.
- C3 *Stability is not validity.* A score that is stable under allowed transforms may still disagree

with how trained readers perceive structural similarity.

- C4 *A scalar does not localize.* Feedback must identify the direction family, the affected region, and whether ink is missing or extra. Fluent text cannot make an incorrect structural attribution reliable.

Figure 1 outlines the framework, from overlapping structural parsing and bounded registration to scalar agreement and localized evidence. Parsing addresses C1, while registration and the score-semantics audit address C2. The human study and diagnostic benchmark evaluate C3 and C4, respectively, through the following research questions.

*RQ1 (representation and generalization).* Can convolutional and transformer parsers trained on the quality-controlled corpus of Sect. 3 recover overlapping direction masks and sparse endpoint masks? How much does performance decline for character identities absent from training?

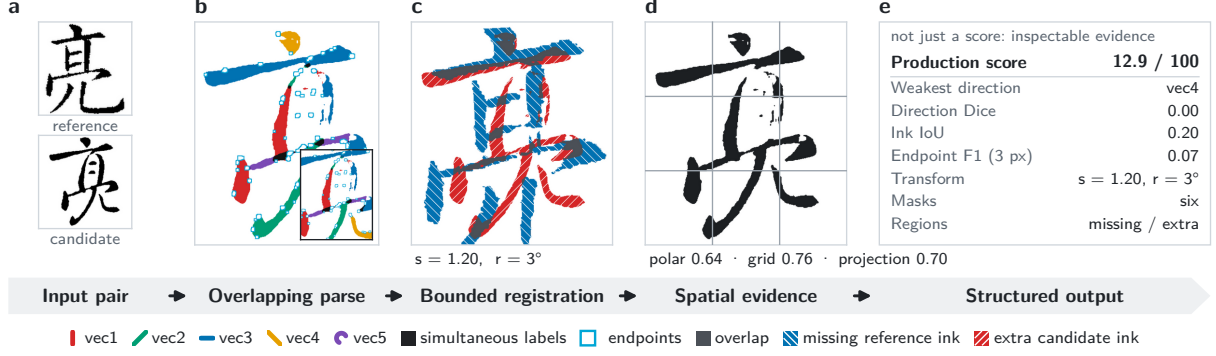
*RQ2 (registration and score behaviour).* Relative to no alignment, does a constrained similarity registration reduce sensitivity to nuisance transforms while preserving penalties for controlled structural changes? Does a wider search improve this trade-off? How do channel-coverage semantics affect the frozen score, and does it distinguish same-character pairs on real references?

*RQ3 (human validity).* How strongly do the frozen production score, the coverage-aware score, and aligned spatial-distribution similarity (ASDS) associate with blinded ratings of same-character structural similarity? Does the ASDS scalar depend on six-channel parsing?

*RQ4 (localized diagnosis).* Which localized diagnostic outputs are reliable on interventions with known ground truth, and why do they fail?

The study makes four contributions aligned with these research questions.

1. For RQ1, we define five independently activated direction-family channels and an endpoint channel, without a per-character prior at inference. The representation differs from Stroke-Seg’s stroke instances (Sect. 2). Semantic quality control supports a matched three-seed comparison of U-Net, DeepLabV3+, and SegFormer-B2 on standard and character-disjoint splits. Ranking depends on target and



**Fig. 1** Reference-conditioned structural assessment on one same-character pair. (a) A regular-script reference and a running-script candidate of the same character are recognizable but structurally different. (b) The candidate is parsed into overlapping direction channels vec1–vec5; the reference follows the same procedure. Black pixels carry multiple direction labels, and cyan outlines mark endpoints. The framed inset shows the multi-label bounding box at reduced scale. (c) A bounded global transform registers the reference to the candidate, compensating global size and tilt (here scale 1.20 and rotation  $3^\circ$ ). The residual comprises overlap (dark grey), missing reference ink (blue, hatched), and extra candidate ink (red, counter-hatched). (d) The aligned silhouette on the  $3 \times 3$  grid used for localized evidence; the values beneath it are the three ASDS spatial-distribution similarities of Sect. 4.4. (e) The structured output includes the production score, weakest direction family and its Dice, ink IoU, endpoint F1, transform, and missing/extra regions. The reported direction Dice belongs to that family, rather than the five-channel mean. All evidence is available before text generation. The cross-style pair is illustrative; the human study uses natural same-character, different-instance pairs from the project corpus. Source images: Calli-Tongji open subset [19], CC BY-NC 4.0.

protocol: DeepLabV3+ leads in-corpus direction parsing, SegFormer-B2 unseen-character direction parsing, and U-Net strict endpoint localization.

- For RQ2, we evaluate constrained similarity registration using controlled nuisance and structural perturbations of 200 real references. Paired comparisons cover no alignment and the tested wider, coarser search, alongside an inactive-channel score audit. Constrained registration substantially reduces nuisance penalties relative to no alignment, while structural edits remain penalized. Residual sensitivity and clipping-conditional summaries delimit this result.
- For RQ3, we compare the three scores in a blinded three-rater study and add a paired direct-ink control. ASDS development is retrospective, with character-grouped internal validation. Its observed association is strongest, but the direct-ink control supports a silhouette interpretation; direction-level evidence provides the rationale for the parser.
- For RQ4, we evaluate localized structured evidence and classify its failures. Required findings are usually surfaced, but exact-cell localization remains unreliable because of multi-cell truth

and direction attribution errors. Language generation remains downstream of the structured evidence.

The intended input is a raster exported from a direct digital writing canvas, not a photograph. The claims concern reference-conditioned structural analysis in this modality. They do not extend to camera robustness, writer-disjoint generalization, stroke-order recovery, historical authenticity, or universal aesthetic grading.

## 2 Related Work

### 2.1 Recognition, document segmentation, and stroke parsing

Large databases such as CASIA-HWDB established benchmarks for online and offline handwritten Chinese character recognition [14]. Recognition assigns character identities; assessing local structure additionally requires spatially resolved outputs. Dense prediction provides a basis for representing the regions that compose a document image. U-Net introduced an encoder–decoder with skip connections [18]; DeepLabV3+ combines atrous spatial pyramids with a decoder [3]; and SegFormer pairs hierarchical transformer

features with a lightweight multilayer-perceptron decoder [24]. In document analysis, dhSegment showed that one generic segmentation formulation can serve many page-level tasks [17]. A recent IJDAR study found that label scarcity, source-target mismatch, and pretraining protocol affect handwritten document segmentation [1]. Standard semantic segmentation assigns each pixel one class [3, 24].

Stroke-Seg addresses stroke-level analysis through a multi-label framework compatible with several segmentation backbones [9]. Its output assigns intersection pixels to multiple strokes, and a prior vector supplies character-specific stroke-composition information. The framework is evaluated by true stroke rate on a brush-calligraphy stroke-segmentation dataset [9].

## 2.2 Computational calligraphy and handwriting-quality evaluation

Early computational calligraphy work represented style with local and global shape features [8, 27]; CalliNet advances style classification with metric learning and triplet supervision [26]. Other systems extract artistic regions relative to a reference image [20] or beautify handwritten glyphs toward a Kai-style template [23]. Designed aesthetic features have also been used to evaluate robotic writing [15]. Large recent resources such as UniCalli support column-level generation and recognition [25].

A survey groups handwritten Chinese character quality evaluation into rule-based geometric measures, template comparison, and learned methods [4]. Defining quality labels and explaining model decisions remain open problems [4]. Siamese networks learn pairwise quality from labelled examples [5], and a recent shared task combined writing-quality grading with comment generation on an expert-annotated dataset [21]. Across these studies, quality spans several constructs, including legibility, geometric correctness, reference similarity, brushwork, and artistic appeal. These constructs can disagree, making the assessment target consequential for its interpretation.

## 2.3 Registration, shape description, and human validation

Image registration ranges from parametric geometric alignment to learned spatial transformers [10, 28]. Registration surveys distinguish rigid, similarity, and affine models from elastic and deformable models by the shape differences they can absorb [28]. Shape contexts describe the spatial distribution of contour points in log-polar bins [2], and spatial pyramids retain coarse layout that an orderless representation would discard [12]. Distributions of this kind can be compared with the bounded, symmetric Jensen-Shannon divergence [13].

Assessment against human judgment requires both a measure of association and an account of rater reliability. Two-way random-effects, absolute-agreement intraclass correlations quantify agreement among raters [11, 16], while bootstrap resampling estimates uncertainty in score-rating associations [7].

## 3 Data Resources

The experiments use a project corpus with manual ground truth (GT), an external reference library, and a human-rating cohort selected from the quality-controlled (QC) project corpus. Table 1 distinguishes their annotation sources and the research questions they support.

### 3.1 Project-collected corpus and six-channel annotation contract

The segmentation corpus was collected in an earlier phase of the project and recovered from its acquisition archive for this study. It was commissioned from contributors described as calligraphy-trained, although no record of individual training survives, who wrote complete characters and, separately, each of their component strokes on a direct digital canvas, marking the two endpoints of every isolated stroke. The six-channel target is composed from these captured components (Fig. 2): a fixed, character-specific mapping assigns each isolated stroke to one of five direction families, and the pixels of strokes in the same family are merged into one mask. Because the strokes were acquired independently, a crossing or adhesion can occupy the same pixel in two families, so

**Table 1** Data resources, their label status, and the research question (RQ) each serves. GT denotes manual six-channel ground truth.

| Resource                         | Origin                                  | Size                                                                | Label status                      | RQ        |
|----------------------------------|-----------------------------------------|---------------------------------------------------------------------|-----------------------------------|-----------|
| Project corpus                   | commissioned direct-digital acquisition | 769 QC-clean of 840 complete; 40 identities                         | GT                                | 1, 3      |
| Standard split                   | frozen sample assignment                | 530/119/120 train/val/test                                          | GT                                | 1         |
| Character-disjoint split         | frozen identity assignment, 28/6/6      | 539/114/116 train/val/test                                          | GT                                | 1         |
| Reference library                | Calli-Tongji Beta, two style categories | 200 images                                                          | model-derived cache, not GT       | mask 2, 4 |
| Perturbation / diagnostic cohort | deterministic edits of cached masks     | 200 references $\times$ 11 perturbations (5 nuisance, 6 structural) | known intervention                | 2, 4      |
| Cross-reference cohort           | reference library                       | 7 positive, 50 negative pairs                                       | pair type only                    | 2         |
| Human-rating cohort              | natural pairs from the project corpus   | 150 pairs + 15 repeats, 3 raters                                    | blinded ratings; development data | 3         |

the composed masks overlap and a crossing pixel legitimately carries more than one direction label. The sixth channel is the union of the manually marked endpoints of all isolated strokes. It encodes stroke termini only, not junctions, turning points, stroke order, or generic salient points. No label in the corpus is produced by any model evaluated in this paper.

The archive records preserve neither a trustworthy mapping from samples to writers nor a standardized measure of contributor expertise, so the paper makes no professional-calligrapher, different-writer, or writer-disjoint claim.

### 3.2 Quality control and split protocols

Recovery of the acquisition archive yielded 894 sample directories across 43 character identities. Complete six-channel ground truth is available for 840 samples from identities 0–39; the 54 samples of identities 40–42 lack the character-specific direction mapping and are excluded rather than pseudo-labelled. A frozen semantic quality-control pass then identified 12 source-image/ground-truth mismatches and 59 non-canonical members of exact image-and-mask duplicate groups. The two exclusion sets do not overlap, leaving 769 unique QC-clean observations of 40 identities (Fig. 3a,b). All exclusion decisions were fixed before the formal model comparison.

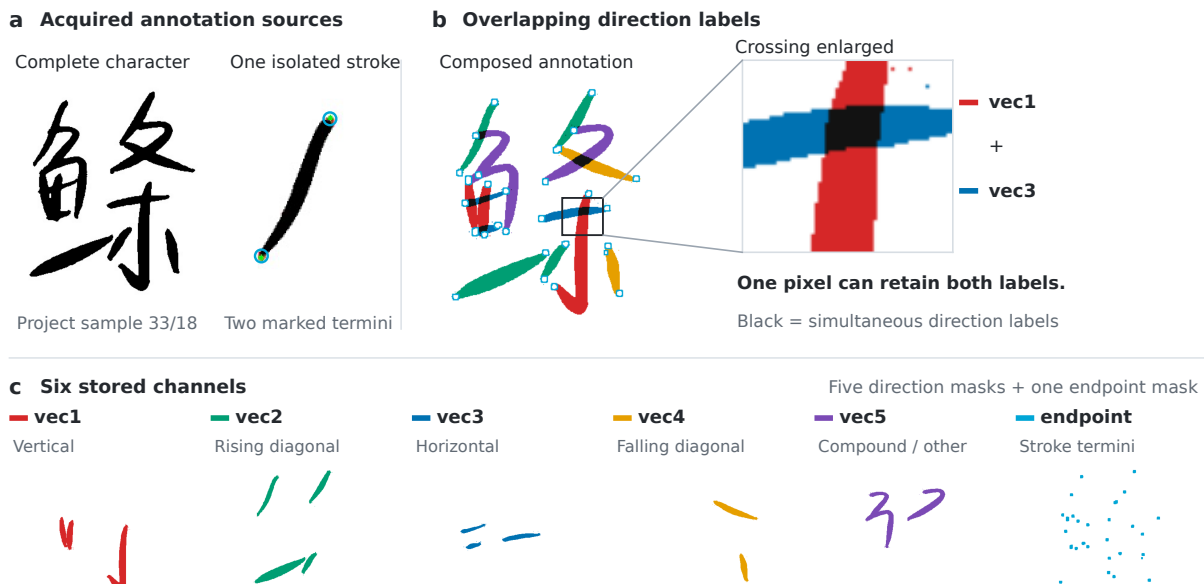
Figure 3 shows the same audit graphically: panel (a) gives the exclusion flow and the retained identity count, panel (b) one deterministic QC-clean observation per retained identity so that

the character coverage of the corpus is visible at a glance, and panel (c) the seven natural cross-style pairs of the separate reference library, whose cached masks are parser outputs rather than annotated ground truth.

Two split protocols are used. The *standard* assignment was frozen before quality control by per-character sample index, with the same index ranges in every identity, and contains 600/120/120 train/validation/test samples before quality control and 530/119/120 after it; it measures in-corporus generalization to new instances of seen identities. The *character-disjoint* assignment was frozen at the identity level before any formal evaluation, with 28/6/6 disjoint identities and 588/126/126 samples before quality control and 539/114/116 after it; it measures transfer to identities never seen in training. No exact-duplicate group crosses either split.

### 3.3 External reference library and derived cohorts

The reference-conditioned experiments use 200 approved images from the publicly downloadable open subset of Calli-Tongji [19] (50 calligrapher-script categories with 100 samples each, released under CC BY-NC 4.0): the 100 images of the Ouyang Xun regular-script category and the 100 images of the Wang Xizhi running-script category. Each reference is parsed once by a frozen SegFormer-B2 checkpoint (version v1, trained with the recipe of Sect. 5.1 on the pre-quality-control standard assignment and frozen before the formal three-seed comparison), and the resulting



**Fig. 2** Six-channel overlapping annotation contract on one project sample (33/18). (a) The complete character and one independently acquired stroke with two manually marked termini; cyan circles are display outlines. (b) The composited annotation and an enlarged crossing of vec1 and vec3. Black pixels retain both direction labels rather than an exclusive class. (c) The five direction channels and the endpoint channel, shown on a shared crop. The vec5 family includes compound, curved, and other non-cardinal strokes; the endpoint channel is the union of manually marked stroke termini, with dilation for display only. The software schema stores the channels as (vec1, ..., vec5, **keypoint**); the last identifier is retained for backward compatibility and denotes this endpoint channel.

six binary masks are cached with their provenance. These cached masks are model-derived reference representations; they are never treated as segmentation ground truth and never enter parser training.

Two cohorts are derived deterministically from the cached masks. The *perturbation cohort* applies the nuisance and structural edits of Sect. 5.2 to every reference, so that registration and score behaviour can be studied without a second parser inference. The *cross-reference cohort* contains all seven natural same-character cross-style pairs that the library supports (Fig. 3c) and 50 different-character negatives selected independently of any model score; the library contains no same-style, different-instance reference pair. The perturbation cohort also serves as the diagnostic cohort of Sect. 5.4, where each known intervention is the ground truth.

### 3.4 Human-rating development cohort

Before any rating was collected, 150 natural same-character, different-instance pairs were selected from the 769 QC-clean project samples, three or

four per identity, excluding mismatches, exact and suspected near duplicates, and repeated instances. Pair roles, asset hashes, blinded identifiers, and display order were frozen, and all automated scores and source identifiers were hidden from the raters.

Three adult raters independently judged structural similarity on a five-point ordinal scale, each rating the 150 canonical pairs plus 15 interspersed hidden repeats. The panel comprised two author-raters with prior calligraphy study, one of whom holds a Level-9 calligraphy certificate, and one non-author rater with master’s-level training in Fine Arts (Calligraphy). The panel is small, domain-qualified, and partly author-involved; it is not an independent expert cohort. The main score comparisons use the annotated masks of both instances, and the paired direct-ink control uses thresholded source rasters. Neither includes parser inference, so these comparisons assess score validity without parser error. The ratings are the ASDS development data of Sect. 5.3, not an external test set.



**Fig. 3** Data resources and quality control. (a) Recovery and exclusion audit: 54 of 894 recovered samples lack a direction mapping, leaving 840 with complete six-channel ground truth. Removing 12 image/GT mismatches and 59 non-canonical exact duplicates yields 769 QC-clean observations of 40 identities. No exact-duplicate group crosses either frozen split. (b) One deterministic QC-clean observation per identity (IDs 0–39). (c) All seven natural same-character cross-style pairs in the external library, represented by cached v1 parser masks, not segmentation ground truth. Source images: Calli-Tongji open subset [19], CC BY-NC 4.0.

## 4 Method

### 4.1 Overview and relation to prior formulations

Let  $I_u$  be a learner image and  $I_r$  an approved reference image of the same target character, one that passed the licence and quality review of the reference library (Sect. 3.3); in deployment the system never scores images of different characters. The system parses both images into overlapping structural masks, registers the reference to the learner with a restricted global transform, and returns (i) six masks per image, (ii) a scalar structural-agreement score, and (iii) localized evidence (Fig. 1). The learner image is called the candidate in the figures and **user** in the software contract; the terms denote the same

image. Three decisions distinguish this formulation from the work of Sect. 2. Unlike Stroke-Seg [9], which recovers individual strokes with a character-specific composition prior supplied as input, the parser predicts five direction families that may overlap plus a separate endpoint channel, uses no per-character prior at inference, and feeds a reference-conditioned comparison rather than a stroke inventory. The comparison is not a learned pairwise similarity network [5]; every score is a fixed, decomposable function of aligned masks whose semantics can be audited. Registration is a bounded similarity transform rather than a learned or deformable warp [10, 28] (Sect. 4.3).

## 4.2 Overlapping six-channel parser

For an RGB character image  $I$ , a parser  $f_\theta$  predicts

$$P = f_\theta(I) \in [0, 1]^{H \times W \times 6},$$

whose channels follow the annotation contract of Sect. 3.1: five direction families and the endpoint channel. Because a crossing pixel may carry several direction labels, each channel is activated by an independent sigmoid rather than by a mutually exclusive softmax; the five direction outputs may therefore overlap, exactly as the composed ground truth does. For direction channel  $c$ , the loss combines class-weighted binary cross entropy (BCE) and soft Dice, averaged over the five channels,

$$\mathcal{L}_{\text{dir}} = \frac{1}{5} \sum_{c=1}^5 \left( \mathcal{L}_{\text{BCE}}^{(c)} + \mathcal{L}_{\text{Dice}}^{(c)} \right).$$

The sparse endpoint target uses focal loss with  $\gamma = 2$  and  $\alpha = 0.75$  plus Dice loss, and a boundary term derived from the masks is weighted by 0.2:

$$\mathcal{L} = \mathcal{L}_{\text{dir}} + \mathcal{L}_{\text{focal, end}} + \mathcal{L}_{\text{Dice, end}} + 0.2 \mathcal{L}_{\text{boundary}}.$$

Binary masks are obtained from  $P$  with per-channel thresholds calibrated on validation data only (Sect. 5.1).

## 4.3 Reference cache and constrained registration

Let  $U$  and  $R$  denote the learner and cached reference binary stacks (Sect. 3.3) and let

$$M(X) = \bigvee_{c=1}^5 X_c$$

be the union of direction ink. For each candidate isotropic scale  $s$  and rotation  $\phi$ , the transformed reference is translated so that its foreground centroid approaches the learner centroid. The production search evaluates nine evenly spaced scales and seven evenly spaced rotations in

$$s \in [0.8, 1.2], \quad \phi \in [-3^\circ, 3^\circ],$$

and selects the candidate maximizing the foreground intersection over union (IoU)

$J(M(U), M(T_{s, \phi, t}(R)))$ . Non-uniform scale, shear, thin-plate splines, and deformable warping are disallowed. This bounded search is a conservative compensation policy: it can absorb page placement and modest global size or rotation differences while leaving local proportion and stroke-region errors available for diagnosis; it implies no invariance within the nominal range, and residual sensitivity is measured in Sect. 6.2.

## 4.4 Structural agreement scores

Three scores are studied. Each is a fixed function of the aligned masks and uses no character identity, writer identity, style identity, or human rating at inference.

**Frozen production score.** After alignment, let  $D_c$  be Dice agreement in direction channel  $c$ ,  $\bar{D} = \frac{1}{5} \sum_{c=1}^5 D_c$ , let  $J_{\text{ink}}$  be union-ink IoU, and let  $F_{\text{end}}^{(3)}$  be endpoint-mask F1 with a three-pixel tolerance. The deployed score is

$$S_{\text{prod}} = 100 \left( 0.55 \bar{D} + 0.25 J_{\text{ink}} + 0.20 F_{\text{end}}^{(3)} \right).$$

It was frozen before the controlled experiments and is never retuned in this paper.

**Coverage-aware score.** The production Dice convention returns 1 for a channel that is empty in both images, so a character that uses fewer direction families can receive credit for structure that is absent. The coverage-aware score keeps the same evidence families and weights but averages Dice only over direction channels active in at least one image, and treats both-empty endpoint masks as unavailable evidence with the remaining weights renormalized. It is reported alongside  $S_{\text{prod}}$  as an audit variant and does not replace it.

**Aligned spatial-distribution similarity (ASDS).** Pixel- and channel-overlap terms say little about how ink mass is organized across the canvas. ASDS therefore compares three spatial distributions of the union silhouette  $M(\cdot)$  after the same frozen registration. *Polar occupancy* centres the foreground at its ink centroid, divides radius by the 95th percentile of foreground radius (clipped to  $[0, 1)$ ), and assigns pixels to four radial and eight angular bins, giving a normalized 32-bin histogram  $h_p$ . Centroid and radius normalization reduce sensitivity to residual translation

and isotropic scale, although rasterization and bin boundaries can still affect the histogram; this component’s invariance is not separately measured here. *Grid occupancy*  $h_g$  counts normalized foreground over a  $3 \times 3$  partition of the canvas and retains coarse location and balance. *Projection profiles*  $h_x$  and  $h_y$  are the normalized column and row sums. For distributions  $p$  and  $q$  with  $m = (p + q)/2$ , the Jensen–Shannon divergence and the Jensen–Shannon similarity (JSS) are

$$D_{\text{JS}}(p, q) = \frac{1}{2} D_{\text{KL}}(p||m) + \frac{1}{2} D_{\text{KL}}(q||m),$$

$$\text{JSS}(p, q) = 1 - \sqrt{D_{\text{JS}}(p, q)},$$

with base-2 logarithms, so that  $\text{JSS} \in [0, 1]$  [13]. With  $s_p = \text{JSS}(h_p(U), h_p(R))$ ,  $s_g = \text{JSS}(h_g(U), h_g(R))$ , and  $s_{\text{proj}}$  the mean of the column and row similarities, the frozen specification is

$$S_{\text{ASDS}} = 100(0.70 s_p + 0.15 s_g + 0.15 s_{\text{proj}}).$$

ASDS was developed after the ratings of Sect. 3.4 were available and its weights were selected on them (Sect. 5.3). Because it consumes only the union silhouette, it does not use the direction semantics of the six-channel masks; whether those masks contribute anything to this scalar beyond their silhouette is tested by the direct-ink control (Sect. 6.3).

## 4.5 Localized evidence and feedback

Alongside the scores, the system preserves the selected transform, the pre-alignment centre distance and ink-area ratio, per-direction Dice, and endpoint agreement. Difference masks between the aligned stacks are summarized by direction channel, by type (reference ink missing from the learner versus extra learner ink), and by a  $3 \times 3$  region set; connected components are exposed as direction-classified regions, not as recovered strokes or stroke order. Deterministic rules rank centre, scale, endpoint, and local-direction findings. An optional text model receives only this structured contract; it may improve fluency but cannot change the score or add brushwork or stroke-order claims the evidence does not contain.

## 5 Experimental Setup

This section states how each research question is tested; the cohorts themselves are defined in Sect. 3.

### 5.1 Parser training and evaluation (RQ1)

Three architectures are compared: a U-Net with 32 base channels, a ResNet-50 DeepLabV3+, and SegFormer-B2. U-Net has no external pre-training; all of its parameters are randomly initialized. DeepLabV3+ loads the torchvision default ImageNet-1K weights for its ResNet-50 encoder, while its atrous spatial pyramid pooling (ASPP) module, low-level projection, decoder, and six-channel output head are randomly initialized. SegFormer-B2 is initialized from `nvidia/segformer-b2-finetuned-ade-512-512`: the compatible Mix Transformer (MiT-B2) encoder and decode-head parameters are loaded from the ADE20K-fine-tuned checkpoint, the incompatible 150-class classifier is replaced by a randomly initialized six-channel classifier, and all layers are then fine-tuned. Each architecture is trained with seeds 20260811, 314159, and 271828 on both split protocols, giving 18 formal runs.

Inputs are letterboxed to  $512 \times 512$  and normalized, and masks are resized by nearest-neighbour interpolation. Batch size is 8 for U-Net and 4 for the two larger models. Label-safe augmentation comprises translation up to 12 pixels, isotropic scale 0.9–1.1, brightness 0.85–1.15, contrast 0.9–1.1, and Gaussian blur with probability 0.15. All models use AdamW with weight decay 0.01, automatic mixed precision, gradient clipping at 1.0, three warm-up epochs, cosine decay to  $10^{-6}$ , at most 120 epochs, and early stopping with patience 15. U-Net uses learning rate  $3 \times 10^{-4}$ ; DeepLabV3+ uses encoder/decoder rates  $10^{-4}/10^{-3}$ ; SegFormer-B2 uses  $3 \times 10^{-5}/3 \times 10^{-4}$ . Positive BCE weights are estimated from at most 200 training samples and capped at 100. Per-channel thresholds are calibrated on the validation split only, and the test split is evaluated once per run.

Direction quality is reported as five-direction Macro Dice and Macro IoU between thresholded predictions and ground truth, and as Boundary

F1, the F1 between the one-pixel morphological boundaries of prediction and ground truth averaged over the five direction channels; endpoint quality is reported as strict endpoint F1, the pixel-exact F1 on the endpoint channel, unlike the three-pixel-tolerant term inside the production score. Test results are summarized as the mean and sample standard deviation over the three seeds. The standard protocol measures in-corpus instance generalization; the character-disjoint protocol measures transfer to unseen characters.

## 5.2 Registration and score-behaviour experiments (RQ2)

Controlled perturbations are applied directly to the 200 cached reference masks. The nuisance family, for which a useful registration should show little sensitivity, comprises translation, rotation, scale up, scale down, and a compound transform. The structural family, for which the score should decrease with severity, comprises terminal deletion, added direction fragment, local fragment shift, direction dilation, direction erosion, and endpoint shift. Each perturbation is applied at four severity levels, except rotation and the compound transform, which use three, so a perturbation with no invalid case contributes 800 rows from the 200 references. The affected channel and axis are selected by stable hashes, never by model error or manual choice. A transformed case that risks canvas clipping is marked invalid with a recorded reason and excluded from valid-score summaries; its fraction is reported. For every perturbation we report valid  $N$ , mean drop with a 95% reference-level bootstrap confidence interval (CI), invalid fraction, and adjacent-severity monotonicity, the fraction of adjacent severity steps at which the score does not increase; median, standard deviation, percentiles, and severity correlations are given in Online Resource 1, Table S1.

The identical reference/perturbation/severity tuples are then scored under three alignment policies: no alignment, the production constrained search of Sect. 4.3, and a wider similarity search with the same number of grid points (nine scales and seven rotations), spanning  $s \in [0.6, 1.4]$  and  $\phi \in [-12^\circ, 12^\circ]$ , fixed in the frozen experiment contract before any result was computed. The wider grid therefore also has coarser spacing (0.10

in scale and  $4^\circ$  in rotation, versus 0.05 and  $1^\circ$  in the production grid); this comparison does not isolate search range from discretization. Comparisons are paired by reference. Confidence intervals resample references; Wilcoxon signed-rank tests [22] and rank-biserial correlation (RBC) effect sizes use per-reference mean paired differences. Family-level comparisons are the primary ablation; perturbation-level results are descriptive and unadjusted (Online Resource 1, Table S2).

Two semantic audits accompany the ablation. The inactive-channel audit counts references with empty direction channels and recomputes the coverage-aware score of Sect. 4.4 on every valid perturbation row (Online Resource 1, Table S3). The cross-reference test scores the seven same-character cross-style pairs and the 50 different-character negatives with  $S_{\text{prod}}$  and summarizes their separation with Cliff’s  $\delta$  [6]; self-pairs are prohibited.

## 5.3 Human-association study and ASDS development (RQ3)

Each score is compared with the three-rater mean on the 150 canonical pairs by Spearman correlation, with 95% intervals from 10,000 bootstrap resamples clustered by target character [7]. The two frozen scores also have nominal, unclustered two-sided Spearman  $p$  values for  $\rho = 0$ ; interpretation emphasizes the character-cluster intervals. Inter-rater reliability uses two-way random-effects, absolute-agreement intraclass correlation coefficients ICC(2,1) and ICC(2,k) [11, 16]. Hidden repeats are excluded from correlation and ICC and are used only for exact agreement, agreement within one point, mean absolute difference (MAD), and quadratic weighted kappa (QWK).

**Post-rating ASDS development.** The original ratings are the ASDS development set. Three fixed distribution families (polar, grid, projection) are considered, and a coarse non-negative simplex search evaluates component weights against the ratings. The search-selected polar weight is retained, and the remaining weight is split equally between grid and projection to reduce sample-specific tuning; the full candidate search is reported in Online Resource 1, Note S5. Five-fold internal validation groups pairs by character identity and selects the weights of each fold on the other characters only, giving a character-grouped

out-of-fold estimate. Character-cluster bootstrap intervals are reported for the frozen score, the out-of-fold predictions, the components, and paired differences from the production and coverage-aware scores; these comparisons are exploratory and not multiplicity-adjusted.

**Paired direct-ink control.** To test whether the ASDS scalar itself requires six-channel masks, ASDS is recomputed on the same 150 pairs with foreground taken directly from the original rasters instead of from the union of the annotated direction masks: grayscale pixels with intensity below 240, the threshold already frozen for dataset quality control, define ink, with no learned pre-processing. Direct-ink masks pass through the same constrained registration and frozen formula; nothing is selected against the ratings for this comparison. We report the Spearman association of the annotated-mask-union variant (the “parsed union” of the frozen artifact) and the direct-ink variant with the ratings, their paired difference, and their mutual rank association, with 10,000 paired resamples clustered by character. Because the control reuses the development pairs, it measures implementation dependence within the same retrospective cohort, not external validity.

## 5.4 Diagnostic evaluation (RQ4)

The diagnostic cohort applies every severity level of the 11 perturbations to the 200 references; the known perturbation of each case supplies deterministic ground truth for the direction channel, the missing/extra type, and the affected  $3 \times 3$  cells. The required finding is the centre-offset finding for translation, the ink-scale finding for scale changes, a local direction finding for the five direction edits, and the endpoint-relation finding for endpoint shifts; rotation and compound cases have no required finding. We report required Recall@3, the fraction of cases whose required finding appears among the top three returned findings, and exact grid localization, which counts a case as correct only when the returned cell equals the truth cell set and is scored on the cases whose perturbation edits a direction channel. Each exact-localization failure is assigned one primary cause in the fixed order local finding absent from the top three, wrong direction channel, wrong missing/extra type, multi-cell or grid-boundary truth, and alignment residual shift; direction and

type errors are characterized only through this classification, and language-model realization is disabled.

# 6 Results and Discussion by Research Question

Results are organized by research question. All reported values come from frozen experiment artifacts; Online Resource 1, Note S6, lists the integrity hashes of those included in the code release.

## 6.1 RQ1 (representation and generalization)

RQ1 is answered by the 18 formal runs of Sect. 5.1 on the standard and character-disjoint splits (Table 2) and by the qualitative comparison of Fig. 4.

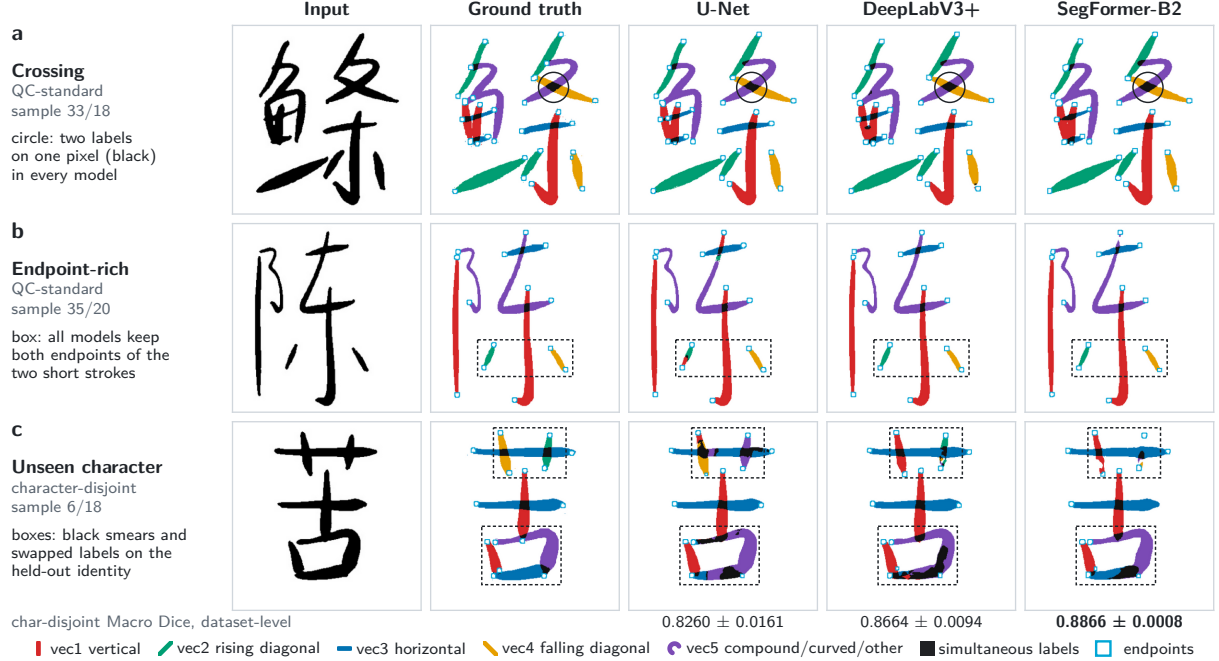
On the standard split, DeepLabV3+ obtains the highest direction Macro Dice ( $0.9630 \pm 0.0006$ ), Macro IoU ( $0.9287 \pm 0.0012$ ), and Boundary F1 ( $0.8443 \pm 0.0028$ ). U-Net is weaker on direction and boundary metrics yet gives the best strict endpoint F1 ( $0.7562 \pm 0.0038$ ). Dense direction segmentation and sparse endpoint localization therefore favour different model configurations.

All models degrade when character identities are held out (Table 2, bottom). SegFormer-B2 gives the strongest character-disjoint direction result: Macro Dice  $0.8866 \pm 0.0008$ , Macro IoU  $0.7977 \pm 0.0013$ , and Boundary F1  $0.6851 \pm 0.0046$ . Its Macro-Dice decrease from the standard split is 7.150 percentage points, versus 9.664 for DeepLabV3+ and 9.360 for U-Net. DeepLabV3+ remains second for direction overlap, and U-Net retains the highest strict endpoint F1 ( $0.6972 \pm 0.0045$ ). Near-saturated in-corpus overlap thus does not imply transfer to unseen character structures, and the best-performing configuration on held-out identities differs from the standard-split leader. These comparisons include the initialization and training choices of Sect. 5.1, rather than isolating architecture alone.

Figure 4 makes the output contract visible: at the crossing in the first row every model emits overlapping labels rather than choosing one family, and the third row shows the errors characteristic of an identity absent from training.

**Table 2** Segmentation results over three fixed random seeds on the QC-clean standard split and the character-disjoint split (train, validation, and test identities disjoint). Values are mean  $\pm$  sample standard deviation; Endpoint F1 is the strict pixel-level score; bold marks the best value in each column within a split.

| Split              | Model        | Macro Dice                            | Macro IoU                             | Endpoint F1                           | Boundary F1                           |
|--------------------|--------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| Standard           | U-Net        | 0.9196 $\pm$ 0.0018                   | 0.8515 $\pm$ 0.0030                   | <b>0.7562 <math>\pm</math> 0.0038</b> | 0.8009 $\pm$ 0.0023                   |
|                    | DeepLabV3+   | <b>0.9630 <math>\pm</math> 0.0006</b> | <b>0.9287 <math>\pm</math> 0.0012</b> | 0.7218 $\pm$ 0.0012                   | <b>0.8443 <math>\pm</math> 0.0028</b> |
|                    | SegFormer-B2 | 0.9581 $\pm$ 0.0031                   | 0.9196 $\pm$ 0.0057                   | 0.7139 $\pm$ 0.0046                   | 0.8259 $\pm$ 0.0088                   |
| Character-disjoint | U-Net        | 0.8260 $\pm$ 0.0161                   | 0.7120 $\pm$ 0.0225                   | <b>0.6972 <math>\pm</math> 0.0045</b> | 0.6430 $\pm$ 0.0136                   |
|                    | DeepLabV3+   | 0.8664 $\pm$ 0.0094                   | 0.7663 $\pm$ 0.0146                   | 0.6755 $\pm$ 0.0031                   | 0.6728 $\pm$ 0.0138                   |
|                    | SegFormer-B2 | <b>0.8866 <math>\pm</math> 0.0008</b> | <b>0.7977 <math>\pm</math> 0.0013</b> | 0.6604 $\pm$ 0.0033                   | <b>0.6851 <math>\pm</math> 0.0046</b> |



**Fig. 4** Frozen qualitative comparison for seed 20260811 on a multi-channel crossing, an endpoint-rich case, and a held-out character (samples are identified as identity/sample index). The first two rows use standard-split models and the third uses character-disjoint models. Call-outs are identical across columns: the circle in (a) marks a crossing where every model emits two labels, the box in (b) two short strokes whose endpoints every model keeps, and the boxes in (c) black smears and swapped labels on the held-out identity. Values under (c) are the dataset-level character-disjoint Macro Dice of Table 2. Colours follow Fig. 2; thresholds are validation-calibrated.

**Answer to RQ1.** The best model configuration depends on the target and split: DeepLabV3+ leads in-corpus direction parsing, SegFormer-B2 unseen-character direction parsing, and U-Net strict endpoint localization. Holding out identities costs 7.150–9.664 Macro-Dice points. Crossing agreement is shown qualitatively, not measured separately; character-disjoint testing measures transfer within one acquisition source, not to unseen writers.

## 6.2 RQ2 (registration and score behaviour)

RQ2 is answered by the controlled-perturbation summary (Table 3), the paired alignment ablation (Table 4, Fig. 5), the inactive-channel audit, and the cross-reference test (Online Resource 1, Table S4) described in Sect. 5.2.

Under the production registration, translation is recovered exactly in all 800 valid cases, whereas residual sensitivity remains for the other transforms: mean penalties are 5.395 points for rotation, 13.653 for scale up, and 16.356 for scale down. Scale up and the compound transform have invalid fractions of 49.0% and 54.0% because of

**Table 3** Controlled perturbation summary. Nuisance rows report absolute score drop; structural rows report signed drop from identity; 95% CIs are reference-level bootstrap intervals; Monotonic is the fraction of non-increasing adjacent severity steps. Full distribution and severity statistics are in Online Resource 1, Table S1.

| Perturbation             | Valid $N$ | Mean drop | 95% CI           | Invalid fraction | Monotonic |
|--------------------------|-----------|-----------|------------------|------------------|-----------|
| Translation              | 800       | 0.000     | [0.000, 0.000]   | 0.000            | 1.000     |
| Rotation                 | 520       | 5.395     | [5.079, 5.747]   | 0.133            | 0.647     |
| Scale up                 | 408       | 13.653    | [12.964, 14.420] | 0.490            | 0.271     |
| Scale down               | 800       | 16.356    | [15.670, 17.074] | 0.000            | 0.500     |
| Compound transform       | 276       | 8.143     | [7.621, 8.717]   | 0.540            | 0.466     |
| Terminal deletion        | 800       | 19.757    | [17.387, 22.087] | 0.000            | 0.992     |
| Added direction fragment | 800       | 11.398    | [9.934, 12.929]  | 0.000            | 0.987     |
| Local fragment shift     | 800       | 4.310     | [3.886, 4.772]   | 0.000            | 0.985     |
| Direction dilation       | 800       | 24.034    | [22.180, 25.955] | 0.000            | 1.000     |
| Direction erosion        | 800       | 22.225    | [20.269, 24.344] | 0.000            | 0.997     |
| Endpoint shift           | 800       | 7.468     | [7.294, 7.641]   | 0.000            | 1.000     |

clipping risk, so their summaries hold only for clipping-safe references and establish no invariance across the library. Among structural perturbations, direction dilation and erosion produce the largest penalties, followed by terminal deletion and added fragments; local fragment shifts and endpoint shifts are weaker but monotonic in severity.

The paired comparison isolates what registration buys. Relative to no alignment, the constrained search reduces the nuisance penalty by 30.118 points (reference-bootstrap 95% CI 29.506–30.727) and increases the penalty for the tested structural interventions by 11.189 points (10.094–12.322); both per-reference signed-rank comparisons are nonzero with rank-biserial effect size 1.0. Without alignment the same structural edits cost only 3.677 points on average against 14.865 under registration (Fig. 5c), so the constrained search does not suppress structural penalties; part of the increase may reflect the centroid-derived translation re-fitting the whole reference after a local edit. The wider, coarser search incurs a 2.896-point higher nuisance penalty (2.757–3.034) while preserving only 0.284 additional points of structural penalty. This tested search configuration does not improve the measured trade-off (perturbation-level results in Online Resource 1, Table S2).

Forty-one of the 200 references contain inactive direction channels (36 with one and five with two; Online Resource 1, Table S3). Both-empty agreement affects 1,252 of the 7,604 valid perturbation rows, and averaging over active channels changes  $S_{\text{prod}}$  by  $-0.289$  points on average and by at most

$-15.402$ : macro overlap can reward absent structure, which motivates the coverage-aware score tested in RQ3.

In the cross-reference test (Online Resource 1, Table S4), the seven same-character cross-style pairs score 20.239 on average (95% CI 15.639–25.069) against 9.741 (8.584–10.970) for the 50 different-character negatives, with Cliff’s  $\delta = 0.84$ . The score carries same-character signal across styles, but the small positive set and the missing same-style condition preclude a broader claim.

**Answer to RQ2.** Constrained registration reduces nuisance penalties while retaining sensitivity to structural edits. The tested wider, coarser search does not improve this trade-off. Residual rotation and scale penalties and clipping exclusions bound the claim. The cross-reference test shows same-character signal, while the inactive-channel audit exposes credit for absent structure, motivating explicit coverage semantics.

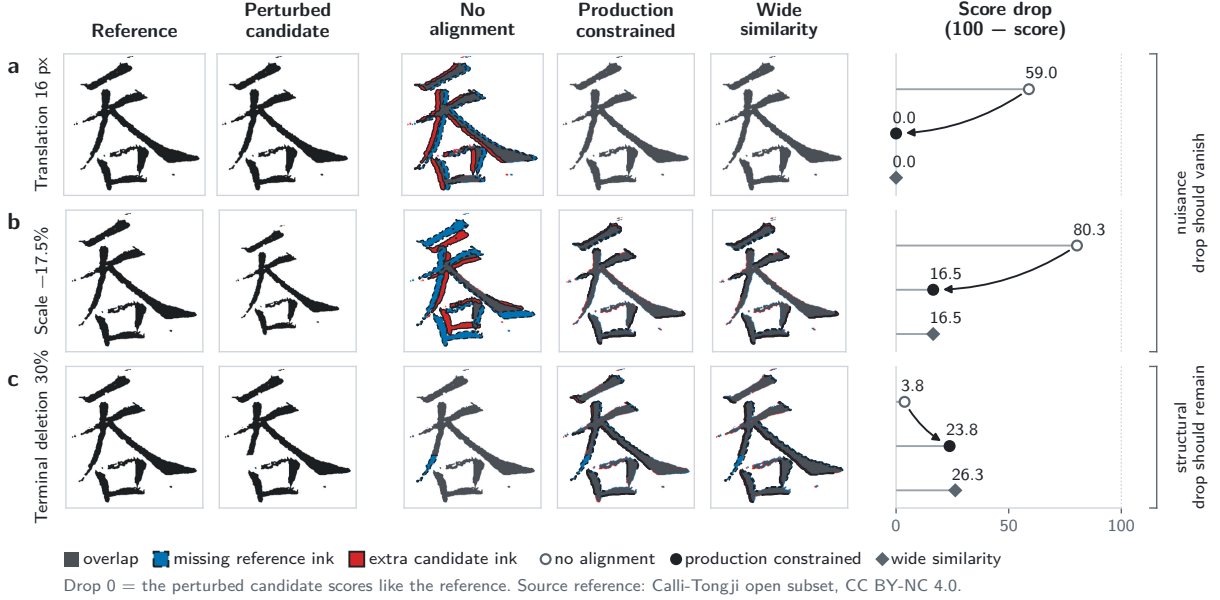
### 6.3 RQ3 (human validity)

RQ3 is answered on the 150-pair human-rating cohort of Sect. 3.4 with the statistics of Sect. 5.3: association of the frozen scores (Table 5), rater reliability (Table 6), ASDS development (Table 7), and the direct-ink control (Table 8, Fig. 6).

The frozen production score correlates modestly with the three-rater mean ( $\rho = 0.297$ , character-cluster 95% CI 0.148–0.441), and the coverage-aware score is stronger ( $\rho = 0.429$ , 0.313–0.535); their paired difference is 0.132 (0.062–0.197). Single-rater reliability is below the 0.50 threshold that Koo and Li [11] label moderate (ICC(2,1) 0.448), whereas the three-rater mean is moderate (ICC(2, $k$ ) 0.709); hidden repeats yield

**Table 4** Paired alignment comparison. Positive benefit differences favour the current constrained alignment and are computed from unrounded means. Ours is the production constrained search of Sect. 4.3; Comparator is no alignment (none) or the wider search (wide); values are mean score drops; RBC is the rank-biserial correlation. Wilcoxon tests use per-reference mean paired differences.

| Comparison | Family     | Refs | Ours   | Comparator | Benefit 95% CI   | $p$     | RBC    |
|------------|------------|------|--------|------------|------------------|---------|--------|
| Ours–none  | nuisance   | 200  | 8.455  | 38.573     | [29.506, 30.727] | < 0.001 | 1.000  |
| Ours–none  | structural | 200  | 14.865 | 3.677      | [10.094, 12.322] | < 0.001 | 1.000  |
| Ours–wide  | nuisance   | 200  | 8.455  | 11.351     | [2.757, 3.034]   | < 0.001 | 1.000  |
| Ours–wide  | structural | 200  | 14.865 | 15.149     | [-0.355, -0.213] | < 0.001 | -0.683 |



**Fig. 5** Alignment ablation on deterministic translation, scale-down, and terminal-deletion cases of one reference. Columns compare no alignment, the production range, and the wider search; dark grey, blue (dashed outline), and red (solid outline) denote overlap, missing reference ink, and extra candidate ink. The right-hand column plots the score drop (100 minus score) of each condition on one 0–100 axis, with an arrow from no alignment to the production range; rows (a)–(b) are nuisance perturbations whose drop should vanish, row (c) a structural edit whose drop should remain. Registration removes the translation penalty in row (a), reduces but does not eliminate the scale penalty in row (b), and retains the structural penalty in row (c). Source reference: Calli-Tongji open subset [19], CC BY-NC 4.0.

**Table 5** Association between the structural scores and the mean rating of three blinded calligraphy-domain raters across 150 canonical pairs. Confidence intervals use 10,000 bootstrap resamples clustered by character;  $p$  values are nominal, unclustered two-sided Spearman tests.

| Score          | $\rho$ | 95% CI         | $p$                   |
|----------------|--------|----------------|-----------------------|
| Production     | 0.297  | [0.148, 0.441] | $2.28 \times 10^{-4}$ |
| Coverage-aware | 0.429  | [0.313, 0.535] | $4.32 \times 10^{-8}$ |

48.9% exact agreement, 93.3% agreement within one point, MAD 0.578, and pooled QWK 0.634.

ASDS reaches  $\rho = 0.556$  (0.454–0.651), and character-grouped out-of-fold weight selection retains  $\rho = 0.540$  (0.432–0.639). The paired intervals for ASDS minus the production and

coverage-aware scores are 0.110–0.410 and 0.017–0.239; the out-of-fold difference from the coverage-aware score is less certain (−0.003–0.224). The polar component is the strongest single term ( $\rho = 0.547$ ); grid and projection reach 0.447 and 0.445. Because the same ratings informed feature design,  $\rho = 0.556$  is a retrospective development result, and  $\rho = 0.540$  is a character-grouped internal-validation estimate that removes weight-selection optimism only: the feature families were also chosen after the ratings, so neither value is an external or prospective confirmation.

The direct-ink control tests whether an annotated-mask union improves ASDS association over thresholded source ink. Annotated-mask-union and direct-ink ASDS reach  $\rho = 0.55623$  and

**Table 6** Calligraphy-domain-rater reliability. ICC uses only the 150 canonical presentations. Repeat statistics use the 15 hidden repeats per rater: Exact and Within  $\pm 1$  are agreement fractions between canonical and repeat ratings, MAD the mean absolute difference, and QWK the quadratic weighted kappa; E01–E03 are blinded rater codes and POOLED aggregates all repeats.

| Rater  | Exact | Within $\pm 1$ | MAD   | QWK   |
|--------|-------|----------------|-------|-------|
| E01    | 0.600 | 1.000          | 0.400 | 0.559 |
| E02    | 0.467 | 1.000          | 0.533 | 0.721 |
| E03    | 0.400 | 0.800          | 0.800 | 0.591 |
| POOLED | 0.489 | 0.933          | 0.578 | 0.634 |

Canonical ICC(2,1)=0.448 and ICC(2,k)=0.709 for 150 pairs and 3 raters.

**Table 7** Development-stage ASDS association with mean human rating (character-cluster bootstrap 95% CI). Grouped OOF: character-grouped out-of-fold weight selection.

| Score or component | Spearman $\rho$ | 95% CI         |
|--------------------|-----------------|----------------|
| Grid only          | 0.447           | [0.303, 0.572] |
| Projection only    | 0.445           | [0.316, 0.564] |
| Polar only         | 0.547           | [0.443, 0.645] |
| ASDS (frozen)      | 0.556           | [0.454, 0.651] |
| ASDS (grouped OOF) | 0.540           | [0.432, 0.639] |

**Table 8** Direct-ink ablation of the frozen ASDS specification on the same 150 development pairs.

| ASDS input                      | Spearman $\rho$ | 95% CI          |
|---------------------------------|-----------------|-----------------|
| Annotated-mask union            | 0.556           | [0.454, 0.651]  |
| Direct ink                      | 0.556           | [0.454, 0.649]  |
| $\Delta\rho$ (annotated–direct) | 0.000           | [-0.010, 0.010] |

Direct ink: grayscale intensity  $< 240$  (frozen QC threshold, Sect. 5.3); same registration, features, and weights for both variants.

0.55583; their difference is  $\Delta\rho = 0.00040$  (95% CI  $-0.01011$ – $0.01030$ ; Table 8 rounds to three decimals), and the two score vectors have rank association 0.9973 (Fig. 6). On these clean direct-digital development pairs, thresholded ink and the annotated-mask union yield closely matched human associations and score rankings. This result supports the silhouette interpretation of ASDS, but does not establish formal equivalence or transfer to degraded input.

**Answer to RQ3.** ASDS has the strongest observed human association, but its internally validated advantage over the coverage-aware score

**Table 9** Primary taxonomy of exact-region diagnostic failures. Fractions are shares of the 3,565 exact-region failures.

| Failure type                    | Count | Fraction |
|---------------------------------|-------|----------|
| Multi-region/boundary truth     | 1922  | 0.539    |
| Wrong direction channel         | 876   | 0.246    |
| Local finding absent from top 3 | 718   | 0.201    |
| Wrong missing/extra type        | 38    | 0.011    |
| Alignment residual shift        | 11    | 0.003    |

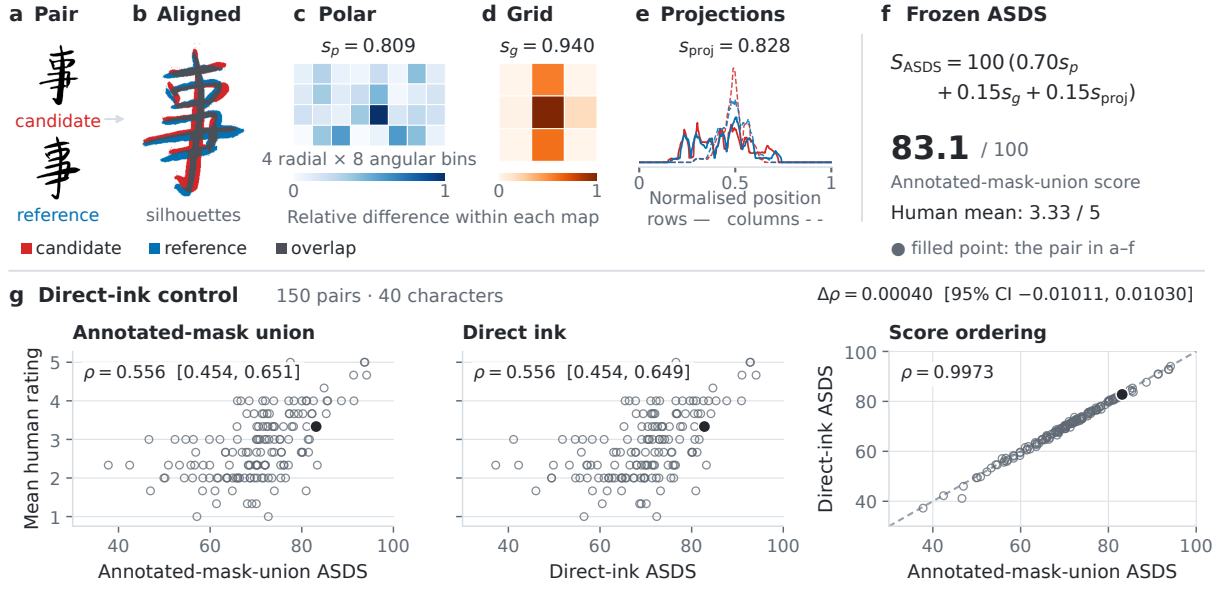
remains unresolved (paired 95% CI  $-0.003$ – $0.224$ ). Its silhouette formulation and the direct-ink control separate scalar agreement from the parser’s role in direction and endpoint evidence. RQ1 evaluates parser accuracy; RQ4 tests the downstream diagnostic mechanism under controlled mask interventions. Human validity remains limited to structural agreement on one retrospective cohort with three raters, two of them authors.

## 6.4 RQ4 (localized diagnosis)

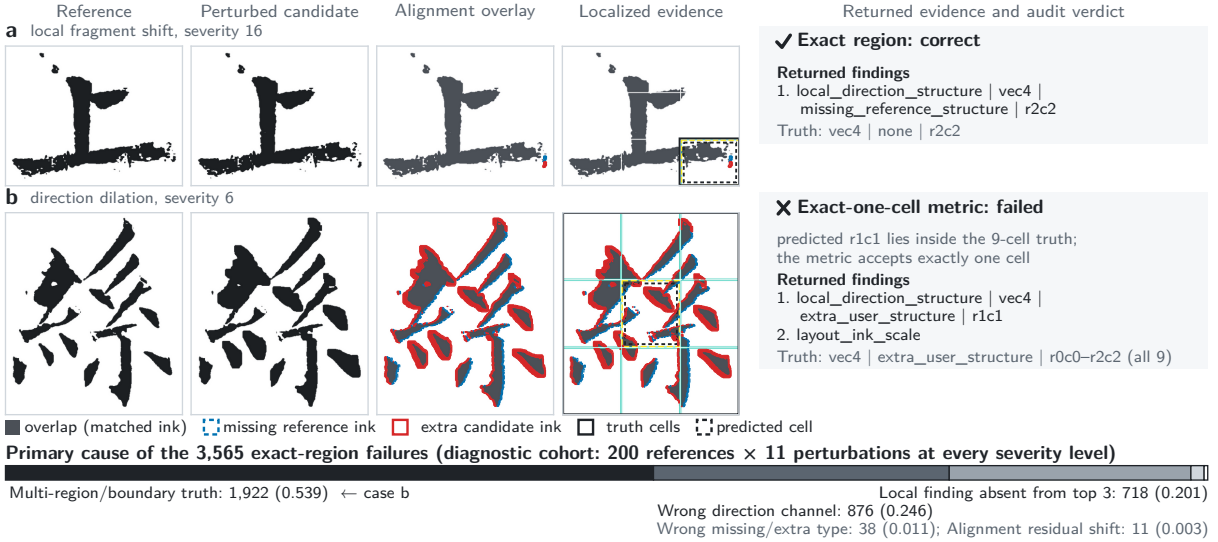
RQ4 is answered on the diagnostic cohort with the metrics of Sect. 5.4 (Table 9, Fig. 7).

Required Recall@3 is 0.701, whereas exact grid localization is 0.109. Under the fixed-order taxonomy, 53.9% of exact-region failures are assigned to multi-cell or grid-boundary truth, 24.6% to wrong direction attribution, and 20.1% to a local finding absent from the top three. Figure 7 shows one exact success and one audited failure of the first kind: the returned cell overlaps the multi-cell change, and the frozen metric still records a failure. The largest assigned category therefore reflects a mismatch between spatial extent and the exact-cell metric, followed by direction attribution. These categories concern structured evidence rather than wording.

**Answer to RQ4.** The evidence layer usually surfaces the required finding but does not reliably assign it to one  $3 \times 3$  cell. Multi-cell truth and direction attribution dominate the assigned failures. Wrong missing/extra type is the primary category in 1.1%, a lower bound under the fixed-order taxonomy. These controlled interventions evaluate the diagnostic mechanism, not teacher agreement or pedagogical usefulness.



**Fig. 6** ASDS mechanism and direct-ink control. (a)–(b) A natural pair and its aligned silhouettes. (c)–(d) Absolute polar (four radial by eight angular bins) and  $3 \times 3$  occupancy differences, each normalized to its own maximum; intensities are not comparable between maps. (e) Projection profiles: rows solid, columns dashed; candidate red, reference blue. (f) Frozen formula and example score. (g) All 150 pairs from 40 identities. Annotated-mask union uses five manual direction masks, not parser predictions. Brackets are character-cluster bootstrap 95% CIs;  $\Delta\rho$  is annotated-mask-union minus direct-ink association. Filled points identify the example; the right-hand dashed line is the identity. Observed associations and rankings are closely matched.



**Fig. 7** Frozen diagnostic evidence: (a) an exact-localization success and (b) multi-cell truth classified as failure by the exact-one-cell metric. Findings report type | direction | missing/extra | cell; rXcY indexes the  $3 \times 3$  grid and user denotes the candidate. Dark grey is overlap, blue (dashed) missing reference ink, and red (solid) extra candidate ink. Solid and dashed black frames mark truth and predicted cells, respectively. Cards state the findings, truth, and audit verdict. The bottom strip summarizes Table 9 over 3,565 exact-region failures; case (b) is in the largest assigned category.

## 7 General Discussion and Limitations

### 7.1 Implications for structural assessment

The four research questions jointly define how structural differences are represented, compensated, scored, and localized. Low-resource parsing must be evaluated on the generalization axis relevant to the application, because the best in-corpus parser did not transfer best to unseen characters. These rankings describe the tested training protocols, including their different initialization schemes. Overlapping labels retain crossing evidence that a mutually exclusive pixel partition cannot express (Fig. 4).

Registration shapes the measurement because the transform family determines which discrepancies remain visible. The bounded search reduces global nuisance penalties while retaining penalties for structural edits. The increase in structural penalties can partly reflect centroid-based refitting after a local edit; it does not independently establish better measurement fidelity. The tested wider search also uses coarser sampling, so its comparison does not isolate transform range. Residual sensitivity prevents a claim of complete invariance under either search.

Scalar agreement and localized diagnostics require different evidence. ASDS depends on the union silhouette, so permuting direction labels leaves it unchanged. Such a change can alter directional comparisons and localized attributions while leaving the separate endpoint mask unchanged. The direct-ink control supports thresholded ink as an input for scalar silhouette comparison in this cohort. Within this framework, the parser supplies direction-specific and endpoint evidence unavailable from the silhouette alone.

The frozen ASDS specification now requires evaluation without modification on new writers, characters, and raters. Diagnostic localization also requires region-overlap or pixel-overlap metrics that credit multi-cell truth.

### 7.2 Limitations

The segmentation corpus contains 40 labelled character identities from one recovered acquisition source. Character-disjoint evaluation does not

establish writer-disjoint generalization (Sect. 3.1), and collection-specific regularities may survive semantic quality control.

Controlled perturbations operate on cached reference masks predicted by the v1 checkpoint, which predates semantic quality control and the formal comparison. Systematic parsing errors in those masks can propagate into RQ2 and RQ4 statistics. Perturbed masks are compared directly without rerunning the parser, so this benchmark does not measure end-to-end performance on perturbed images. Scale-up and compound summaries are conditional on clipping-safe references because their invalid fractions are high. The external library supports only seven same-character cross-style pairs, limiting conclusions about discrimination across styles.

The human cohort contains 150 pairs rated by three domain-qualified raters, two of whom are authors. Single-rater reliability is poor, while reliability of the three-rater mean is moderate. The primary score comparison uses annotated masks, so it excludes deployment parser error; the direct-ink control bypasses parsing. ASDS feature design followed inspection of the ratings, making its association retrospective and its character-grouped validation internal. The direct-ink control reuses this cohort and provides no independent confirmation.

The contribution of registration to ASDS was not isolated. Its polar term is centroid- and scale-normalized, so part of the observed association may persist without alignment. Agreement at multi-label crossing pixels is not measured separately, and diagnostic ground truth comes from engineered interventions rather than teacher judgments. The supported modality remains clean direct digital writing; robustness to camera capture has not been evaluated (Sect. 1).

## 8 Conclusion

We presented an auditable framework for assessing a handwritten Chinese character against a same-character reference under limited labelled data. Overlapping parsing, constrained registration, and human-audited comparison provide a shared basis for scalar agreement and localized evidence.

For RQ1, three parsers learn overlapping direction masks and a separate endpoint mask from a 769-sample quality-controlled corpus.

DeepLabV3+ leads standard-split direction parsing, SegFormer-B2 leads character-disjoint direction parsing, and U-Net leads strict endpoint localization. For RQ2, constrained similarity registration reduces nuisance penalties by 30.118 points relative to no alignment while increasing structural penalties by 11.189 points. The tested wider, coarser search does not improve this trade-off, and the reported benefit remains conditional on clipping-safe references.

For RQ3, blinded ratings from three domain-qualified raters, including two authors, associate most strongly with ASDS, followed by the coverage-aware and frozen production scores. ASDS reaches  $\rho = 0.556$  in retrospective development and  $\rho = 0.540$  under character-grouped internal validation. Its internally validated advantage over the coverage-aware score remains unresolved. Direct-ink and annotated-mask-union ASDS show nearly identical association in this cohort. The parser supplies direction, overlap, endpoint, and localized evidence beyond this scalar comparison. For RQ4, controlled mask interventions show that the evidence layer usually surfaces the required finding but does not reliably localize it to one cell. Multi-cell or grid-boundary truth is the largest assigned failure category (53.9%) under the fixed-order taxonomy and exact-one-cell metric.

Together, these results support inspectable, reference-conditioned structural assessment of clean direct-digital writing, with structured evidence established before language generation. Independent human validation and improved localization assessment remain necessary before broader claims about aesthetic or pedagogical usefulness.

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## Statements and Declarations

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(Entrepreneurship Training category), “One-Stroke: A Vision-Model-Based Intelligent Feedback System for Chinese Calligraphy”. No external funding channel is recorded, and no grant number was available at the time of writing.

**Competing Interests.** The authors declare no competing interests.

**Ethics Approval.** Not applicable. The study analysed images of handwritten characters and perceptual similarity ratings; it involved no clinical, medical, or behavioural intervention on human subjects. The handwriting corpus was commissioned by the project without recording writer identities, and the structural-similarity ratings were provided by two of the authors and one collaborating calligraphy specialist as part of the study design, stored under evaluator codes, with no sensitive personal information collected.

**Informed Consent.** Not applicable. No identifiable personal data of writers or raters are reported; the three raters took part in the rating study voluntarily as contributors to the project and were informed that the task assessed character structure similarity, not their own ability.

**Data Availability.** The frozen data contracts and result tables that support the reported values are in the code repository named under Code Availability: the quality-control audit and exclusion contract and both split assignments (`artifacts/data_qc`, `artifacts/data_audit`, `artifacts/paper_ijdar/character_disjoint`); the per-seed segmentation results and check-point manifest (`artifacts/paper_ijdar/task1`); the ASDS development features, character-grouped folds, frozen specification, and report (`artifacts/paper_ijdar/spatial_score_development`); the 150-pair direct-ink ASDS table (`artifacts/paper_ijdar/direct_ink_asds`); the reference-library manifest with per-image source paths and SHA-256 digests (`references/calli_tongji_beta_manifest.csv`); and the figure provenance manifest. Online Resource 1, Note S6, lists the digests of every frozen artifact. The external reference images are the Ouyang Xun regular-script and Wang Xizhi running-script categories (100 images each) of the Calli-Tongji open subset [19], publicly downloadable from ModelScope under CC BY-NC 4.0 at <https://www.modelscope.cn/datasets/CalliTo>

ngji/Calli-Tongji\_A\_Dataset\_of\_Historical\_Calligraphy\_Styles; they are reproduced in Figs. 1, 3, and 5 with attribution and are not redistributed. The project-collected handwriting images and six-channel masks, the controlled-perturbation, alignment-ablation, inactive-channel, and cross-reference result files, and the blinded 150-pair rating files (evaluator codes only) are available from the corresponding author on reasonable request for non-commercial research use.

**Code Availability.** Source code, frozen experiment contracts, analysis scripts, and non-restricted reproducibility artifacts are maintained at <https://github.com/LiuXiaofan-0321/OneStroke2026>; an identity-free archival copy will be supplied if anonymous review is required.

**Author Contributions.** X.L.: Conceptualization, methodology, software, formal analysis, visualization, writing—original draft, project administration. R.Z.: Data curation, methodology, software, validation, writing—review and editing. Y.F.: Software, system integration, validation, writing—review and editing. All authors contributed to the investigation, approved the submitted manuscript, and accept responsibility for the work.

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